BEU CONNECT

Bihar Engineering University, Patna

End Semester Examination - 2022

Time: 03 Hours

Code: 101606 ***Subject: Geotechnical Engineering - II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven Question only): **[14]**

(a) The compressibility of clays may also be caused by the following factor.
(i) Changes in salinity
(ii) Ion exchange
(iii) Expulsion of double layer water from between the grains
(iv) Swelling of clay

(b) Bishop's simplified method of slices satisfies
(i) Only the moments equilibrium
(ii) Only the vertical forces equilibrium
(iii) Only the horizontal forces equilibrium
(iv) All the statics equations, except the horizontal forces equilibrium

(c) Identify the incorrect statement. The stability of a slope is decreased by
(i) Removal of a part of slope by excavation
(ii) Shock caused by an earthquake
(iii) Pore water pressure in the soil
(iv) Providing a berm at the toe

(d) For saturated, normally consolidated soils, skempton's pore pressure coefficients can be represented as
(i) A < 1, *B*=1
(ii) A > 1, *B* > 1
(iii) A > 1, *B* < 1
(iv) A < 1, *B*> 1

(e) In an Undrained triaxial compression test, the sample failed at a deviator stress of 200 kN/m² when the cell pressure was 100 kN/m². The cohesion intercept is
(i) 200 kN/m²
(ii) 100 kN/m²
(iii) 300 kN/m²
(iv) 50 kN/m²

(f) For an undisturbed sample, the area ratio of the samples should be
(i) Zero
(ii) 10% or less
(iii) 10% to 20%
(iv) More than 20%

(g) If the actual value of the standard penetration number (N) is greater than 15 for fine sands below water table, the corrected value of N is
(i) 15 +([N+15] / [2])
(ii) 15 −([N+15] / [2])
(iii) 15 +([N−15] / [2])
(iv) 15 +([15−N] / [2])

(h) Taylor's stability number curves are used for the analysis of stability of slopes. The angle of shearing resistance used in the chart is the
(i) Effective angle
(ii) Apparent angle
(iii) Mobilized angle
(iv) Weighted angle

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(i) After the hydrostatic pressure is reduced to zero, some compression of soil takes place by the process of

(i) Effective pressure
(ii) Load increment
(iii) Primary consolidation
(iv) Secondary consolidation

(j) The Area Ratio for the soil sampler used for soil exploration is given by:

(i) $[D_2^2 - D_1^2] / [D_1^2] \times 100$
(ii) $[D_2^3 - D_1^3] / [D_1^3] \times 100$
(iii) $[D_2 - D_1] / [D_1] \times 100$
(iv) $[D_1 - D_2] / [D_2] \times 100$

Q.2 Solve both questions : **[14]**

(a) Explain the Friction circle method for the analysis of stability of finite slopes.

(b) A granular soil having saturated unit weight = 19 kN/m³ and angle of internal frictional resistance = 35°. A slope has to be made of this granular soil. Determine the safe angle of slope considering the factor of safety = 1.3 if: (i) The soil slope is dry without seepage (ii) The seepage occurs parallel to the soil slope with the water table at a depth of 1.5 m from ground level.

Q.3 Solve both questions : **[14]**

(a) Define consolidation. Explain different types of consolidation.

(b) What are the assumptions of Terzaghi's consolidation theory? Derive the Terzaghi's One-dimensional consolidation equation.

Q.4 Solve both questions : **[14]**

(a) What are the causes of sample disturbance? How are they taken care of in samples for undisturbed sampling?

(b) A direct shear test is run on medium sand under the normal stress of 50 kPa. The maximum shear stress at failure is measured as 30 kPa. Draw the Mohr's circle at failure, and determine the magnitude of the principle stresses in the failure zone. What is the orientation to the plane of maximum shear stress at failure.

Q.5 Solve this question : **[14]**

Explain the conditions when vane shear test is suitable for determining the shear strength of the soil. Derive an expression for shear strength of soil in terms of applied torque and vane dimensions considering shearing at top, bottom and surrounding soil. An in situ vane shear test was conducted in a clay soil at the bottom of a borehole. A torque of 150 N-m was required to shear the soil. What was the undrained shear strength of clay neglecting shearing resistance at the top? The vane was 100 mm in diameter and 150 mm long.

Q.6 Solve this question : **[14]**

In the laboratory test on a clay sample, of thickness 25 mm drained at top only, 50% consolidation occurred in 11 minutes. Find the time required for the corresponding clay layer in the field, 2 m thick and drained at top and bottom to undergo 70% consolidation. Assume T_{50}= 0.197, T_{70}= 0.405.

Q.7 Solve both questions : **[14]**

(a) A sample of dry cohesionless soil whose angle of internal friction is 35°, is subjected to a triaxial test. If the minor principal stress (σ_3) is 105 kPa, at what values of deviator stress ($\Delta\sigma$) and major principal stress (σ_1) will the test specimen fail?

(b) Explain Swedish circle method for analysis of stability of finite slope.

Q.8 Solve both questions : **[14]**

(a) What is coefficient of consolidation of soil? What are the methods to determine coefficient of consolidation. Describe any one.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(b) Describe the salient features of a good sub-soil investigation report.

Q.9 Write short notes on any four of the following: **[14]**

(a) Wash boring
(b) Split-spoon samplers (with neat sketches)
(c) Static-cone penetration test
(d) Taylor stability number
(e) Effective shear strength parameters
(f) Pre-consolidated and under consolidated soils

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store